

PROJECT DOCUMENTATION

Project Title

Rising Waters

Project Subtitle

AI-Powered Flood Prediction System using Machine Learning

Abstract

Rising Waters is an AI-powered flood prediction system that predicts flood risk using rainfall and weather parameters. The project uses Machine Learning algorithms to analyze environmental data and provide accurate flood predictions. The application is developed using Python and Streamlit, offering a simple, fast, and user-friendly interface for disaster management.

Objectives

- Predict flood risk accurately.
 - Analyze rainfall and weather data.
 - Provide early flood warnings.
 - Develop an easy-to-use web application.
 - Improve disaster preparedness.
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Software Requirements

- Python 3.x
 - Streamlit
 - Pandas
 - NumPy
 - Scikit-learn
 - XGBoost
 - Joblib
 - Git
 - VS Code
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Hardware Requirements

- Intel Core i3 or above
- 4 GB RAM or higher
- 10 GB Free Storage

- Internet Connection
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Dataset

The project uses historical rainfall and weather data collected from open-source datasets. The dataset contains rainfall, temperature, humidity, and other environmental parameters required for flood prediction.

Machine Learning Model

The XGBoost algorithm is used for training the flood prediction model because of its high accuracy and efficiency.

Application Workflow

1. User enters rainfall and weather parameters.
 2. Input data is validated.
 3. The trained model processes the data.
 4. Flood risk prediction is generated.
 5. Result is displayed on the Streamlit dashboard.
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Project Features

- AI-based flood prediction
 - Interactive Streamlit dashboard
 - Fast prediction
 - User-friendly interface
 - High prediction accuracy
 - GitHub version control
 - Streamlit deployment
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Results

The application successfully predicts flood risk with high accuracy and provides quick responses through an interactive web interface.

Future Enhancements

- Real-time weather API integration

- SMS and Email alert system
 - Mobile application
 - Interactive flood risk maps
 - Live rainfall monitoring
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Conclusion

The Rising Waters project demonstrates the effective use of Machine Learning for flood prediction. The application is reliable, accurate, easy to use, and can support disaster management authorities in making timely decisions.